

Recent human evolution

Recent human evolution refers to evolutionary adaptation, sexual and natural selection, and genetic drift within *Homo sapiens* populations, since their separation and dispersal in the Middle Paleolithic about 50,000 years ago. Contrary to popular belief, not only are humans still evolving, their evolution since the dawn of agriculture is faster than ever before.^{[1][2][3][4]} It has been proposed that human culture acts as a selective force in human evolution and has accelerated it,^[5] however, this is disputed.^{[6][7]} With a sufficiently large data set and modern research methods, scientists can study the changes in the frequency of an allele occurring in a tiny subset of the population over a single lifetime, the shortest meaningful time scale in evolution.^[8] Comparing a given gene with that of other species enables geneticists to determine whether it is rapidly evolving in humans alone. For example, while human DNA is on average 98% identical to chimpanzee DNA, the so-called Human Accelerated Region 1 (**HAR1**), involved in the development of the brain, is only 85% similar.^[2]

Following the peopling of Africa some 130,000 years ago, and the recent Out-of-Africa expansion some 70,000 to 50,000 years ago, some sub-populations of *Homo sapiens* have been geographically isolated for tens of thousands of years prior to the early modern Age of Discovery. Combined with archaic admixture, this has resulted in relatively significant genetic variation. Selection pressures were especially severe for populations affected by the Last Glacial Maximum (LGM) in Eurasia, and for sedentary farming populations since the Neolithic, or New Stone Age.^[9]

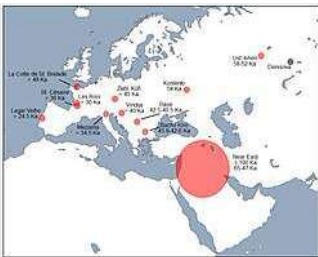
Single nucleotide polymorphisms (SNP, pronounced 'snip'), or mutations of a single genetic code "letter" in an allele that spread across a population, in functional parts of the genome can potentially modify virtually any conceivable trait, from height and eye color to susceptibility to diabetes and schizophrenia. While approximately 2% of the human genome codes for proteins and a slightly larger fraction is involved in gene regulatory function. If the environment remains stable, the beneficial mutations will spread throughout the local population over many generations. A dominant trait. An extremely beneficial allele could become ubiquitous in a population in as little as a few centuries whereas typically take millennia.^[10]

Human traits that emerged recently include the ability to free-dive for long periods of time,^[11] adaptations for living concentrations are low,^[2] resistance to contagious diseases (such as malaria),^[12] light skin,^[13] blue eyes,^[14] lactase persistence weaning),^{[15][16]} lower blood pressure and cholesterol levels,^{[17][18]} retention of the median artery,^[19] reduced prevalence susceptibility to diabetes,^[20] genetic longevity,^[20] shrinking brain sizes,^{[21][22]} and changes in the timing of menarche and men-

The **Neolithic Revolution**, also known as the **First Agricultural Revolution**, was the wide-scale transition of many human cultures during the Neolithic period from the egalitarian lifestyle of nomadic and semi-nomadic hunting and gathering to one of agriculture and domesticated animals used for food and labor.

Archaic admixture

Genetic evidence suggests that a species dubbed *Homo heidelbergensis* is the last common ancestor of Neanderthals, Denisovans, and *Homo sapiens*. This common ancestor lived between 600,000 and 750,000 years ago, likely in either Europe or Africa. Members of this species migrated throughout Europe, the Middle East, and Africa and became the Neanderthals in Western Asia and Europe while another group moved further east and evolved into the Denisovans, named after the Denisova Cave in Russia where the first known fossils of them were discovered. In Africa, members of this group eventually became anatomically modern humans. Migrations and geographical isolation notwithstanding, the three descendant groups of *Homo heidelbergensis* later met and interbred.^[24]



Map of western Eurasia showing areas and estimated dates of possible Neandertal–modern human hybridization (in red) based on fossil samples from indicated sites^[25]

Archaeological research suggests that as prehistoric humans swept across Europe 45,000 years ago, Neanderthals went extinct. Even so, there is evidence of interbreeding between the two groups as humans expanded their presence in the continent. While prehistoric humans carried 3–6% Neanderthal DNA, modern humans have only about 2%. This seems to suggest selection against Neanderthal-derived traits.^[26] For example, the neighborhood of the gene *FOXP2*, affecting speech and language, shows no signs of Neanderthal inheritance.

Introgression of genetic variants acquired by Neanderthal admixture has different distributions in Europeans and East Asians, pointing to differences in selective pressures.^[28] Though East Asians inherit more Neanderthal DNA than Europeans;^[27] East Asians, South Asians, Australo-Melanesians, Native Americans, and Europeans all share Neanderthal DNA, so hybridization likely occurred between Neanderthals and their common ancestors coming out of Africa.^[29] Their differences also suggest separate hybridization events for the ancestors of East Asians and other Eurasians.^[27]

Following the genome sequencing of three Vindija Neanderthals, a draft sequence of the Neanderthal genome was published and revealed that Neanderthals shared more alleles with Eurasian populations—such as French, Han Chinese, and Papua New Guinean—than with sub-Saharan African populations, such as Yoruba and San. According to the authors of the study, the observed excess of genetic similarity is best explained by recent gene flow from Neanderthals to modern humans after the migration out of Africa.^[30] But gene flow did not go one way. The fact that some of the ancestors of modern humans in Europe migrated back into Africa means that modern Africans also carry some genetic materials from Neanderthals. In particular, Africans share 7.2% Neanderthal DNA with Europeans but only 2% with East Asians.^[29]

Some climatic adaptations, such as high-altitude adaptation in humans, are thought to have been acquired by archaic admixture. An ethnic group known as the Sherpas from Nepal is believed to have inherited an allele called EPAS1, which allows them to breathe easily at high altitudes, from the Denisovans.^[24] A 2014 study reported that Neanderthal-derived variants found in East Asian populations showed clustering in functional groups related to immune and haematopoietic pathways, while European populations showed clustering in functional groups related to the lipid catabolic process.^[note 1] A 2017 study found correlation of Neanderthal admixture in modern European populations with traits such as skin tone, hair color, height, sleeping patterns, mood and smoking addiction.^[31] A 2020 study of Africans unveiled Neanderthal haplotypes, or alleles that tend to be inherited together, linked to immunity and ultraviolet sensitivity.^[29]

